**20** The table shows some values of $x$ and $y$ that satisfy the equation $y = a\cos x° + b$

| $x$ | 0 | 30 | 60 | 90 | 120 | 150 | 180 |
|---|---|---|---|---|---|---|---|
| $y$ | 3 | $1 + \sqrt{3}$ | 2 | 1 | 0 | $1 - \sqrt{3}$ | $-1$ |

Find the value of $y$ when $x = 45$

$\cos 0° = 1$ and $\cos 90° = 0$.

At $x = 90$: $1 = a \times 0 + b$, so $b = 1$.

At $x = 0$: $3 = a \times 1 + 1$, so $a = 2$.

$\cos 45° = \dfrac{\sqrt{2}}{2}$

$y = 2 \times \dfrac{\sqrt{2}}{2} + 1 = 1 + \sqrt{2}$

**1 mark**
Use exact trigonometric values

**1 mark**
Find b from a suitable row

**1 mark**
Find both constants

**1 mark**
Correct exact value

$1 + \sqrt{2}$

**(Total for Question 20 is 4 marks)**